



Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
Ano letivo 2026/2027

Practical Exercise no. 1

Theme *Intro to Data Science & Python/Scikit-learn*

Exercise The aim of this exercise is to prepare a functional Python virtual environment to start learning, practising, and developing Machine Learning software as part of the Dados e Aprendizagem Automática curricular unit.

Tasks Given this statement, you should follow the following steps:

T1. Download and install the Miniconda package for your operating system (Windows, macOS, Linux). The package is available at <https://www.anaconda.com/docs/getting-started/miniconda/system-requirements>;

T2. Start the Miniconda prompt and create a Python3.14 virtual environment. The following libraries must be installed in this environment:

- Numpy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyterlab

T3. Activate the created virtual environment and check the installed libraries. Validate the installation of the libraries shown in T2;

T4. Briefly check the documentation of each library mentioned in question T2. Identify their relevance in the context of developing Machine Learning algorithms.